

3.11 :-

X: Random-purchase of defectives = 3, $n = 7$, $D = 2$.

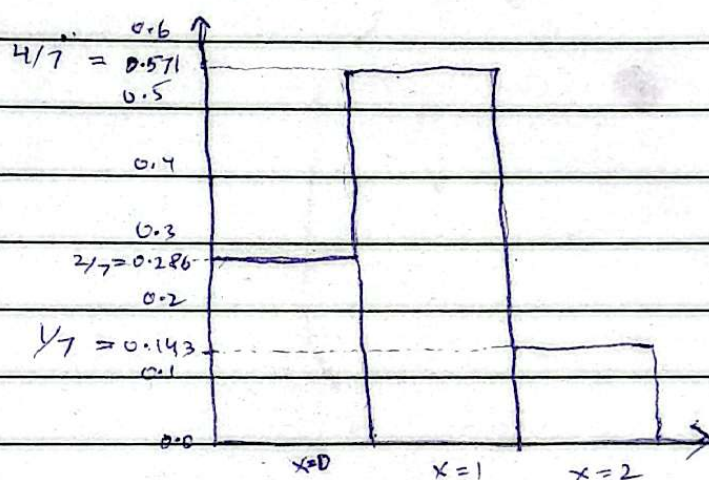
$$P(X=x) = \frac{{}^2C_x ({}^5C_{3-x})}{{}^7C_3}$$

$$P(X=0) = \frac{{}^2C_0 ({}^5C_3)}{{}^7C_3} = 2/7$$

$$P(X=1) = \frac{{}^2C_1 ({}^5C_2)}{{}^7C_3} = 4/7$$

$$P(X=2) = \frac{{}^2C_2 ({}^5C_1)}{{}^7C_3} = 1/7$$

X	P(X)
0	2/7
1	4/7
2	1/7



3.18 :-

$$f(x) = 2(1+x)/27 \quad ; \quad 2 \leq x \leq 5$$

$$\int f(x) dx = \frac{2}{27} \int (1+x) dx = \frac{x^1}{27} \cdot \frac{(1+x)^2}{x^1} + C \Rightarrow (1+x)^2/27 + C$$

$$(a) P(X < 4) = F(4) - F(2) = \left(\frac{1+4}{27}\right)^2 - \left(\frac{1+2}{27}\right)^2 = 16/27$$

$$(b) P(3 \leq X \leq 4) = F(4) - F(3) = \frac{25}{27} - \frac{16}{27} = 9/27 = 1/3$$

3.20 :-

Find pdf, find F(x), evaluate $P(3 \leq X \leq 4)$

$$f(x) = \begin{cases} \frac{2(1+x)}{27} & 2 \leq x \leq 5 \\ 0 & \text{elsewhere} \end{cases}$$

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F(x) CDF

$$F(x) = \int_2^x \frac{2(1+t)}{27} dt = \frac{2}{27} \left[t + \frac{t^2}{2} \right]_2^x$$

$$\Rightarrow \frac{2}{27} \left[\left(x + \frac{x^2}{2} \right) - \left(2 + 2 \right) \right] = \frac{2}{27} \left[\frac{2x + x^2 - 8}{2} \right]$$

$$F(x) \Rightarrow \frac{x^2 + 2x - 8}{27} ; F(x) = \begin{cases} 0 & x < 2 \\ \frac{x^2 + 2x - 8}{27} & 2 \leq x \leq 5 \\ 1 & x > 5 \end{cases}$$

For $P(3 \leq x \leq 4) :-$

$$F(4) - F(3) = \frac{4^2 + 2(4) - 8}{27} - \frac{3^2 + 2(3) - 8}{27}$$

$$= \frac{16 - 7}{27} = \frac{1}{3}$$

3.13 :-

x	0	1	2	3	4
f(x)	0.41	0.57	0.16	0.05	0.01
F(x)	0.41	0.78	0.94	0.99	1.00

3.39 :-

(a)

$P(X,Y) = \{ X: \# \text{ of Oranges} \}$
 $Y: \# \text{ of Apples} \}$

$$P(X,Y) = \frac{{}^3C_x {}^2C_y {}^3C_{4-x-y}}{{}^8C_4}$$

X \ Y	0	1	2
0	0	3/70	3/70
1	3/70	18/70	4/70
2	9/70	19/70	8/70
3	3/70	1/70	0

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(b)

$$P[(X,Y) \in A], A = \{(x,y) | x+y \leq 2\}$$

$$A = P(0,0), P(0,1), P(1,1), P(1,0), P(2,0), P(0,2)$$

$$P[(X,Y) \in A] = \frac{0}{70} + \frac{3}{70} + \frac{18}{70} + \frac{3}{70} + \frac{9}{70} + \frac{3}{70} = \frac{36}{70}$$

3.49 :-

(a) X	1	2	3
g(x)	0.1	0.35	0.55

(b) Y	1	3	5
h(y)	0.2	0.5	0.3

$$(c) P(Y=3 | X=2) = \frac{f(2,3)}{g(2)} = \frac{0.1}{0.35} = 0.286$$

Covariance :

$$\text{Cov}(x,y) = E(x,y) - E(x)E(y)$$

$$E(x,y) = \sum xy \cdot f(x,y)$$

$$E(1,y) = (1)(1)(0.05) + (1)(3)(0.05) + (1)(5)(0) = 0.2$$

$$E(2,y) = (2)(1)(0.05) + (2)(3)(0.1) + (2)(5)(0.2) = 2.7$$

$$E(3,y) = (3)(1)(0.1) + (3)(3)(0.35) + (3)(5)(0.1) = 4.95 + 7.85$$

$$E(x,y) = 7.85$$

x	g(x)	E(x)	y	h(y)	E(y)
1	0.1	0.1	1	0.2	0.2
2	0.35	0.7	3	0.5	1.5
3	0.55	1.65 +	5	0.3	1.5
		2.55			3.2

$$\text{Cov}(x,y) = 7.85 - (2.55)(3.2) = -0.31$$

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3.57 :-

$$f(x, y, z) = \begin{cases} kxy^2z & x > 0, y < 1, 0 < z < 2, \\ 0 & \text{elsewhere} \end{cases}$$

(a) find 'k'.

$$k \int_0^2 \int_0^1 \int_0^1 xy^2z \, dx \, dy \, dz = 1$$

$$k \int_0^2 \int_0^1 \left[\frac{x^2 y^2 z}{2} \right]_0^1 dy \, dz \Rightarrow \frac{k}{2} \int_0^2 \int_0^1 y^2 z \, dy \, dz$$

$$\frac{k}{2} \int_0^2 \left[\frac{y^3 z}{3} \right]_0^1 dz \Rightarrow \frac{k}{6} \int_0^2 z \, dz$$

$$\frac{k}{12} \left[\frac{z^2}{2} \right]_0^2 \Rightarrow \frac{k}{12} \cdot 2 = 1 \Rightarrow \boxed{k = 3}$$

$$(b) \int_0^2 \int_{1/2}^1 \int_0^1 xy^2z \, dx \, dy \, dz$$

3.62 :-

(a)	x	1	3	5	7
	P(x)	0.4	0.2	0.2	0.2

$$(b) P(4 < X \leq 7) = P(7) - P(4) = 1 - 0.6 = \boxed{0.4}$$

4.1 :-

$$E(x) = x \cdot P(x)$$

$$= 0(0.41) + 1(0.57) + 2(0.16) + 3(0.05) + 4(0.01)$$

$$E(x) = 0.88$$

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4.7 :-

$$\text{Expected gain} = 4000 \times 0.3 - 1000 \times 0.7$$

$$= 1200 - 700$$

$$E(x) = \$500$$

4.12 :-

$$\text{r.v.'s pdf} = \int x \cdot f(x) \, dx$$

$$= \int_0^1 2x(1-x) \, dx \Rightarrow 2 \int_0^1 x - x^2 \, dx$$

$$\Rightarrow 2 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 \Rightarrow 2 \left[\left(\frac{1}{2} - \frac{1}{3} \right) - 0 \right] = \frac{1}{3}$$

$$\text{Avg profit} = \frac{1}{3} \times 5000 = \$1667$$

per automobile.

4.20 :-

$$\text{Let } Y = g(X), \text{ then } E(Y) = \int_0^\infty g(x) e^{-x} \, dx, \text{ where } g(x) = e^{2x/3}$$

$$E(Y) \Rightarrow \int_0^\infty e^{2x/3} \cdot e^{-x} \, dx \Rightarrow E(Y) = \int_0^\infty e^{-x/3} \, dx$$

$$\Rightarrow \left[-3 e^{-x/3} \right]_0^\infty \Rightarrow \frac{-3}{e^{1/3}} - \frac{-3}{e^{0/3}} \Rightarrow \boxed{E(Y) = 3}$$

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4.34:-

$$S.D = \sqrt{\sigma^2}, \quad \sigma^2 = \text{Var}(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = -2^2(0.3) + 3^2(0.2) + 5^2(0.5)$$

$$= 1.2 + 1.8 + 12.5 = 15.5$$

$$E(X) = -2(0.3) + 3(0.2) + 5(0.5) = 2.5$$

$$\sigma^2 = 15.5 - 2.5^2 = 9.25 \Rightarrow \sigma = 3.04$$

4.58:-

$$Y = g(x) \quad \dots Y \text{ is a function of } x$$

$$E(Y) = \int_{-\infty}^{\infty} g(x)f(x) dx$$

$$E(Y) = \int_0^1 (60x^2 + 39x)(x) dx + \int_1^2 (60x^2 + 39x)(2-x) dx$$

$$E(Y) = 28 + 81 = 109 \text{ kWh}$$

5.12:-

$$q.p = 0.75, n = 9, p = 0.25, P(X < 4) = ?$$

$$P(X=0) = {}^9C_0 (0.75)^9 (0.25)^0 = 0.0751$$

$$P(X=1) = {}^9C_1 (0.25)^1 (0.75)^8 = 0.2253$$

$$P(X=2) = {}^9C_2 (0.25)^2 (0.75)^7 = 0.3003$$

$$P(X=3) = {}^9C_3 (0.25)^3 (0.75)^6 = 0.2336 + 0.834$$

$$P(X < 4) = 83.4\%$$

5.26:-

$$P(X=6) = \binom{8}{6} \left(\frac{6}{10}\right)^6 \left(1 - \frac{6}{10}\right)^2 = 0.209 = 20.9\%$$

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5.57:-

$$(a) \lambda = 2, P(X=k) = \frac{(e^{-\lambda})^k}{k!}$$

$$P(X \geq 4) = 1 - P(X \leq 3)$$

$$P(0) = (e^{-2}) \left(\frac{2^0}{0!}\right) = e^{-2}$$

$$P(1) = (e^{-2}) \left(\frac{2^1}{1!}\right) = 2e^{-2}$$

$$P(2) = (e^{-2}) \left(\frac{2^2}{2!}\right) = 2e^{-2}$$

$$P(3) = (e^{-2}) \left(\frac{2^3}{3!}\right) = \frac{4}{3}e^{-2}$$

$$P(X \geq 4) = 1 - [e^{-2} + 2e^{-2} + 2e^{-2} + \frac{4}{3}e^{-2}] = 0.143$$

$$(b) P(X=0) = e^{-2} = 0.135$$

6.8:-

$$(a) x = 17, z = \frac{x - \mu}{\sigma}, \mu = 30, \sigma = 6; z = \frac{17 - 30}{6} = -2.16$$

$$P(X > 17) = 1 - P(Z < -2.16)$$

$$= 1 - 0.0154 = 0.9846$$

$$(b) z = \frac{22 - 30}{6} = -1.33, P(X < 22) = P(Z < -1.33) = 0.0918$$

$$(c) P(32 < X < 41) = P(X < 41) - P(X < 32)$$

$$z_1 = \frac{32 - 30}{6} = 0.33 \quad P(Z < 0.33) = 0.6293$$

$$z_2 = \frac{41 - 30}{6} = 1.83 \quad P(Z < 1.83) = 0.9664$$

$$P(32 < X < 41) = 0.9664 - 0.6293 = 0.3371$$

$$(d) \phi = 0.8 (80\%) \rightarrow z = 0.84, x = \mu + z\sigma \Rightarrow x = 30 + (0.84)6$$

$$x = 35.04$$

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6.11 (e) $P(z < z_1) = 0.1250$

$$z_1 = -1.15$$

$$P(z > z_1) = 1 - 0.125 = 0.8750$$

$$z_2 = 1.15$$

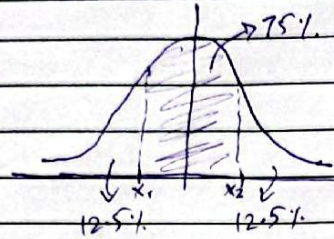
$$\therefore x_1 = \mu + z_1 \sigma$$

$$x_1 = 30 + (-1.15)6$$

$$x_1 = 25.1$$

$$x_2 = 30 + (1.15)6$$

$$x_2 = 36.9$$



6.11 :-

(a) $\mu = 200$, $\sigma = 15$, $x = 224$, $z = \frac{x - \mu}{\sigma} = \frac{224 - 200}{15} = 1.6$

$$P(z > 1.6) = 1 - P(z < 1.6)$$

$$= 1 - 0.9452 = 0.0548 = 5.48\%$$

(b) $P(191 < x < 209) = P(x < 209) - P(x < 191)$

$$z_1 = \frac{191 - 200}{15} = -0.60 ; \phi(-0.60) = 0.2743$$

$$z_2 = \frac{209 - 200}{15} = 0.60 ; \phi(0.6) = 0.7257$$

$$P(z < 209) - P(z < 191) = 0.7257 - 0.2743 = 0.4514 = 45.14\%$$

(c) Probability of Overflow: $P(x > 230)$

$$P(x > 230) = 1 - P(x < 230)$$

$$= 1 - P(z < 2.00)$$

$$z = \frac{230 - 200}{15} = 2.00$$

$$P(x > 230) = 1 - 0.9772 = 0.0228$$

$$1000 \times 0.0228 = 22.8 \approx 22 \text{ cups will overflow out of 1000 cups.}$$

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(d) $\phi = 0.25 \rightarrow z = -0.67$, $x = 7$, $\mu = 200$, $\sigma = 15$

$$x = \mu + \sigma z = 200 + (15)(-0.67)$$

$$x \approx 189.95 \text{ mL}$$

6.21 :- full question not given

6.21 :-

$$\mu = 10000, \sigma = 100, \text{continuity correction} = \pm 25$$

(a) $P(x > 10150 + 25) = P(x > 10175)$

$$P(z > 1.75) = 1 - P(z < 1.75)$$

$$z = \frac{10175 - 10000}{100}$$

$$P(x > 10150) = 1 - 0.9599 = 0.0401 = 4.01\%$$

$$z = 1.75$$

(b) $P(x < 9800 + x > 10200)$

$$= 1 - P(9800 < x < 10200)$$

$$= 1 - [P(x < 10200) - P(x < 9800)]$$

$$= 1 - [P(x < 10225) - P(x < 9775)]$$

$$= 1 - [P(z < 2.25) - P(z < -2.25)]$$

$$1 - (0.9898 - 0.0122)$$

$$1 - 0.9776 = 0.0224$$

$$P(\text{scrap}) = 0.0224 = 2.24\%$$

$$z_1 = \frac{9775 - 10000}{100}$$

$$= -2.25$$

$$z_2 = \frac{10225 - 10000}{100}$$

$$z_2 = 2.25$$